01 — Timeline

Health Hub Business Plan v2 | Comprehensive Timeline and Phase Execution Plan Version: 2.0 | **Date:** March 2026

> *This document maps the full journey from prototype to production across all 21 scenario-investment combinations. It incorporates the v2 parameter set: a 12-person part-time team (3 core + 7 developers + 2 advisors), 745 hours of remaining web work, a fixed 480-hour Android APK contract, 750 hours of parallel operational workstreams, and an employed operations staff ladder that begins at Pilot 1. All assumptions trace to* `params.md`.

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1. Timeline Philosophy

Three principles govern this timeline. They have not changed from the earlier planning cycle; v2 sharpens the numbers behind them without altering the logic.

Principle 1 — The Phase Order Stays Intact

Pre-Pilot, Pilot 1, Pilot 2, Production Readiness. This sequence is non-negotiable. Each phase exists because it validates a distinct class of risk:

Phase	Risk Class Resolved
Pre-Pilot	Technical stability, internal rehearsal, scope lock
Pilot 1	Live-user viability, callback discipline, first partner activation
Pilot 2	Scale readiness, staffing ladder, revenue learning, market expansion prep
Production Readiness	Compliance, full-stack hardening, launch governance, support schedule

No amount of funding or team capacity can safely skip a phase. The phases can compress in duration, but they cannot be reordered or eliminated.

Principle 2 — Lower Coding Effort Does Not Erase Operating Work

The v2 parameter set reduced the web platform estimate from v1's 682-1,413 hour three-point range to a tighter 720-768 hour band (modeled at 745 hours). That is a real correction based on a real team estimate. But the company must still build:

- A patient-facing Android APK (480 hours, contracted)
- A callback-enabled operating layer with trained doctors and admin staff
- Doctor training protocols, admin SOPs, escalation trees, and partner kits
- Release governance, compliance documentation, and production support systems

These parallel workstreams total 750 hours of effort (per params.md Section 17). They consume founder and founding-team time directly, which means the engineering calendar is not the only constraint on the timeline. Even if the web platform were finished tomorrow, the company would not be ready to launch.

Principle 3 — The APK Is a Parallel Critical Path

The Android APK is not a "later nice-to-have." In the East African market context (35-45% urban smartphone penetration in Ethiopia, mobile-first user behavior), the APK is the primary patient access channel. The 480-hour APK workstream runs on a dedicated 3-4 person sub-team and must hit two milestones:

APK Milestone	Target Phase	Scope
Core patient journeys functional	Pilot 1	Registration, dashboard, consultation request, basic video/chat, notifications
Full patient scope complete	Pilot 2	Prescription viewing, lab results, payments, full video/audio/chat, offline resilience

The APK schedule constrains the Pilot 1 gate independently of web progress. If the APK slips, Pilot 1 cannot proceed even if the web platform is ready, because the primary patient interface would be missing.

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2. Shared Parallel Workstreams

These workstreams run alongside software development throughout the entire build cycle. They are not optional add-ons; they are the difference between shipping a product and launching a healthcare operation.

2.1 Workstream Overview

#	Parallel Track	Starts	Ends	Est. Hours	Primary Owner	Why It Matters
W1	Doctor training protocols and playbooks	Pre-Pilot	Production Readiness	120-160	Founding Member 1 + Medical Advisor	Care quality cannot depend on informal verbal handoff; employed doctors (starting Pilot 1) need structured onboarding and clinical guidelines for tele-consultation
W2	Admin callback SOPs and escalation trees	Pre-Pilot	Production Readiness	80-120	Founding Member 1	Travel, diagnostics, pharmacy, and exception cases remain partly manual; admin staff (starting Pilot 1) need documented decision paths for every callback type

W3	Admin operational guidelines (daily workflows)	Pre-Pilot	Pilot 2	60-80	Founding Member 1	Day-to-day workflows (shift handoff, queue management, partner escalation) must be written before admin headcount scales beyond 2
W4	Technical support runbooks	Pilot 1	Production Readiness	60-80	Technical Head	Live users create real issue-response obligations; the dev team cannot absorb L1/L2 support and still build
W5	Partner onboarding kits (clinics, labs, pharmacies)	Pre-Pilot	Pilot 2	80-120	Owner + Founding Member 1	Clinics, labs, and pharmacies need operational materials, agreements, and integration guides — not just login credentials
W6	Release governance and rollback checklists	Pre-Pilot	Production Readiness	40-60	Technical Head	Essential for a part-time team where no single person monitors releases 24/7; rollback must be mechanical, not heroic
W7	Staffing calibration and refresher training	Pilot 1	Production Readiness	40-60	Founding Member 1	Coverage quality must grow with user volume; calibration cycles review shift adequacy, callback response times, and doctor utilization
W8	Launch analytics and reporting setup	Pilot 2	Production Readiness	60-80	Technical Head	Investor-grade reporting, operational dashboards, and KPI tracking require dedicated setup beyond code instrumentation
W9	Compliance and regulatory documentation	Pre-Pilot	Production Readiness	80-120	Owner + Advisors	Ethiopia and Kenya have distinct regulatory environments; documentation must be prepared before each market entry

Total parallel operational effort: 620-880 hours (modeled at **750 hours** expected case per params.md)

2.2 Workstream Phasing Chart

This chart shows which workstreams are active in each phase. Shaded cells indicate active effort.

Workstream	Pre-Pilot	Pilot 1	Pilot 2	Prod Readiness
W1: Doctor training	DRAFT	DELIVER + ITERATE	REFRESH	FINALIZE
W2: Admin callback SOPs	DRAFT	DELIVER + ITERATE	REFRESH	FINALIZE
W3: Admin daily workflows	DRAFT	DELIVER	FINALIZE	—
W4: Tech support runbooks	—	DRAFT + DELIVER	ITERATE	FINALIZE
W5: Partner onboarding kits	DRAFT	DELIVER BATCH 1	DELIVER BATCH 2	—
W6: Release governance	DRAFT v1	ITERATE	ITERATE	FINALIZE
W7: Staffing calibration	—	INITIAL BASELINE	CALIBRATE	FINALIZE
W8: Analytics and reporting	—	—	BUILD + ITERATE	FINALIZE
W9: Compliance and regulatory	RESEARCH	ETHIOPIA PREP	KENYA PREP	CLOSEOUT

2.3 Founder Time Allocation Impact

These workstreams compete for the same founder and founding-team hours that also go toward investor relations, product decisions, partner meetings, and strategic oversight. At Level 1, where effective capacity is ~320 hours/month, the operational workstreams consume a disproportionate share of available time, which is the primary driver of timeline stretch at lower investment levels.

Investment Level	Technical Hours/Mo	Ops Workstream Hours/Mo (est.)	Remaining for Strategy/BD
L1 (Bootstrapped)	~320	~40-60	~30-50
L2 (Ideal)	~480	~50-70	~50-80
L3 (Fully Funded)	~640	~60-80	~80-120

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3. Phase-by-Phase Objectives and Internal Gates

3.1 Pre-Pilot

Dimension	Detail
Timing band	M0 to M3 (L3) through M0 to M6 (L1), depending on team capacity
User scale	100-200 internal users (team members, advisors, selected friends/family)
Technical focus	Architecture freeze. Bug triage from audit (19 issues, all resolved). Role-flow completion for GP, specialist, pharmacy, diagnostics, admin. APK backlog lock and sprint 0. Core security hardening (auth, SSR safety, rate limiting). Web platform: ~200-280 hours of the 745-hour budget consumed here. APK: ~100-160 hours of the 480-hour budget consumed here (project setup, core scaffolding, auth flow, basic navigation).
Operating focus	Draft doctor training playbooks (W1). Draft admin callback SOPs and escalation trees (W2). Draft admin daily workflow guidelines (W3). Draft partner onboarding kit templates (W5). Release governance v1 (W6). Begin compliance research for Ethiopia (W9). Internal rehearsal loops: 2 cycles minimum, simulating real patient flows with team members acting as patients, doctors, and admin.
Commercial focus	Warm-network partner grooming (1-3 clinics in Addis Ababa identified and relationship-building started). Demo readiness for investor conversations. Early pricing rehearsal using Ethiopia benchmarks (ETB 150-300 GP consult, ETB 300-600 specialist). No revenue expected ($0/mo per params.md).
Infrastructure	Tech infra: $65-130/mo (hosting, DB, minimal video). Ops infra: $0-20/mo. One-time: $200-300 for initial admin phones (2 devices). No employed ops staff yet.

Internal Minor Checkpoints:

Checkpoint	Timing (relative to phase)	Criteria
CP-1: Architecture freeze	Week 2-3	All major technical decisions documented; no further framework/library changes
CP-2: Rehearsal cycle 1	40% through phase	One complete patient journey (registration through consultation) tested end-to-end by team
CP-3: Issue burn-down check	60% through phase	All P0 and P1 audit issues verified closed; no regression
CP-4: Rehearsal cycle 2	80% through phase	All six role flows (patient, GP, specialist, pharmacy, diagnostics, admin) tested with realistic scenarios
CP-5: Runbook v1 signoff	90% through phase	Doctor playbook draft, admin SOP draft, and release governance v1 reviewed by founding team

Gate Criteria to Advance to Pilot 1:

- [ ] Core web role flows stable (GP queue, specialist referral, pharmacy lookup, diagnostics tab navigation, admin user management)
- [ ] APK scope locked and sprint plan confirmed with Android team
- [ ] APK core scaffolding complete (auth, navigation, basic dashboard)
- [ ] 2 internal rehearsal cycles passed with documented findings and fixes
- [ ] Doctor training playbook v1 drafted and reviewed
- [ ] Admin callback SOP v1 drafted and reviewed
- [ ] Partner onboarding kit template ready
- [ ] At least 1 partner clinic relationship warm and ready for Pilot 1 activation
- [ ] Release governance checklist v1 in place
- [ ] Ethiopia regulatory research summary completed
- [ ] No unresolved P0 or P1 technical issues

3.2 Pilot 1

Dimension	Detail
Timing band	2-3 months duration
User scale	Up to 1,000 users (first external users in Ethiopia)
Technical focus	Stable provider web platform. Core patient APK journey functional (registration, dashboard, consultation request, basic video/chat, notifications). Incident handling and monitoring. Callback visibility in admin dashboard. Web platform: ~150-200 hours consumed. APK: ~200-240 hours consumed (core patient journeys, video integration, notification system).
Operating focus	**Employed staff activate:** 2 doctors + 2 admin ops + 2 tech support (monthly cost: ~$2,482 per params.md). Live callback operations begin. Doctor training delivered and first iteration based on real cases (W1). Admin SOPs delivered and iterated with real callback data (W2). Admin daily workflows delivered (W3). Tech support runbooks drafted and delivered (W4). Partner onboarding batch 1 executed (W5). Staffing baseline established (W7).
Commercial focus	First real partner usage (1-3 clinics active). First paid consultation experiments at subsidized pricing ($0-500/mo revenue expected per params.md). Live service feedback collection. Partner satisfaction baseline.
Infrastructure	Tech infra: $160-300/mo. Ops infra: $110-220/mo (phone top-ups, VoIP, support ticketing). One-time: $400-600 for laptops/workstations, $200-300 misc office setup.

Internal Minor Checkpoints:

Checkpoint	Timing (relative to phase)	Criteria
CP-6: Partner onboarding batch 1	Week 1-2	At least 1 clinic fully onboarded, staff trained, able to receive referrals
CP-7: First live consultations	Week 2-3	At least 5 real patient consultations completed end-to-end
CP-8: Callback backlog review	Mid-phase	Callback queue metrics reviewed; average response time under 4 hours
CP-9: Doctor training refresh	60% through phase	Training updated based on first 2-4 weeks of real case patterns
CP-10: First paid consults	70% through phase	At least 1 paid consultation (even at subsidized rate) successfully processed
CP-11: APK core milestone check	End of phase	APK core patient journeys functional and tested with real users

Gate Criteria to Advance to Pilot 2:

- [ ] Live Ethiopia pilot ran for minimum 1 month (2 months preferred) with real patients
- [ ] Measurable throughput: at least 50 consultations completed
- [ ] Callback discipline demonstrated: documented SOP followed, response times tracked
- [ ] Escalation handling tested: at least 3 escalation scenarios handled per documented procedure
- [ ] Doctor utilization tracked and reported (target: >30% during operating hours)
- [ ] APK core patient journeys confirmed functional with real users
- [ ] Tech support handled live issues without requiring dev team intervention for L1/L2
- [ ] Partner feedback collected and documented from all active clinics
- [ ] No unresolved critical (P0) technical issues
- [ ] Revenue experiment data collected (even if revenue is minimal)

3.3 Pilot 2

Dimension	Detail

Timing band	3-13 months depending on scenario and investment level
User scale	5,000-10,000 users
Technical focus	Full APK scope complete (prescription viewing, lab results, payments, full video/audio/chat, offline resilience). Web analytics and partner reporting dashboards. Partial automation of manual processes. Release process tightening. Performance optimization. Web platform: ~200-265 hours consumed (remainder of 745 budget, including hardening and analytics). APK: ~120-180 hours consumed (remaining scope, polish, payment integration). QA/DevOps/security: ~120-180 hours of the 220-hour budget consumed.
Operating focus	**Staffing ladder deepens:** 4 doctors + 4 admin ops + 2 tech support (monthly cost: ~$4,361 per params.md). Admin daily workflow guidelines finalized (W3). Tech support runbooks iterated (W4). Partner onboarding batch 2 (W5). Staffing calibration cycle 1 (W7). Launch analytics setup begins (W8). Kenya regulatory prep (W9). Reporting loops established: weekly ops review, monthly staffing review.
Commercial focus	Larger user acquisition push via targeted marketing in Addis Ababa. Stronger partner proof points (5+ clinics active). Revenue at 50% of target pricing ($500-3,000/mo per params.md). Kenya market prep package assembled. Investor reporting pack v1 if external capital is active.
Infrastructure	Tech infra: $300-600/mo. Ops infra: $160-320/mo (call center software, expanded devices). One-time: $300-600 for additional admin phones/tablets.

Internal Minor Checkpoints:

Checkpoint	Timing (relative to phase)	Criteria
CP-12: Reporting pack v1	20% through phase	Operational and financial reporting template delivered to founding team and investors (if applicable)
CP-13: Staffing ladder activation	30% through phase	Doctor and admin headcount increased per ladder; shift coverage confirmed
CP-14: APK full scope milestone	40% through phase	All patient-facing APK features functional and in testing
CP-15: Payment/process review	50% through phase	Payment flow end-to-end tested; process automation candidates identified
CP-16: Kenya prep package	70% through phase	Regulatory summary, partner targets, market entry plan documented
CP-17: Staffing calibration cycle 1	80% through phase	Shift adequacy, callback response times, doctor utilization reviewed and adjustments made
CP-18: Scale readiness assessment	90% through phase	Platform load tested for 10,000 concurrent users; ops team confirmed ready

Gate Criteria to Advance to Production Readiness:

- [ ] 5,000-10,000 user-capable operation exists with documented support processes
- [ ] Reporting established: weekly ops metrics, monthly financial summary
- [ ] Service consistency measurable: callback response SLA met >80% of the time
- [ ] Doctor utilization >50% during operating hours
- [ ] Full APK scope complete and stable
- [ ] Web platform hardening complete (all 745 hours consumed or remaining items de-risked)
- [ ] Kenya market entry plan documented with regulatory pathway identified
- [ ] Monthly active patients >1,000 (secondary success metric per params.md)
- [ ] Revenue learning sufficient to validate pricing corridor
- [ ] No unresolved P0 or P1 technical issues
- [ ] Staffing calibration completed with documented findings

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3.4 Production Readiness

Dimension	Detail
Timing band	4-12 months depending on path
User scale	Transitioning to full scale
Technical focus	Final hardening pass. Monitoring and alerting maturity. Release freeze window. Production support schedule formalized. Security audit closeout. Performance benchmarks documented. Remaining QA/DevOps budget consumed (~40-100 hours).
Operating focus	**Full staffing ladder:** 6 doctors + 6 admin ops + 3 tech support (monthly cost: ~$6,542 per params.md). Doctor training finalized (W1). Admin SOPs finalized (W2). Tech support runbooks finalized (W4). Release governance finalized (W6). Staffing calibration finalized (W7). Analytics and reporting finalized (W8). Compliance closeout for Ethiopia; Kenya compliance in progress (W9). Launch playbooks written and rehearsed. Support schedule with shift coverage confirmed. Fallback operations documented (what happens if a system goes down).
Commercial focus	Production pricing posture established (full pricing corridors per params.md). Partner maturity: contracts formalized, SLAs in place. Launch communication readiness (marketing materials, partner announcements). Revenue target: $3,000-15,000/mo per params.md. LTV:CAC tracking initiated (target: >3x per params.md).
Infrastructure	Tech infra: $600-1,100/mo. Ops infra: $200-400/mo. Moving toward production steady-state: $800-1,500/mo combined.

Internal Minor Checkpoints:

Checkpoint	Timing (relative to phase)	Criteria
CP-19: Launch playbook signoff	25% through phase	Complete launch playbook reviewed and approved by all founding team members
CP-20: Support coverage signoff	40% through phase	2-shift or 3-shift coverage confirmed; all staff trained and scheduled
CP-21: Security audit closeout	50% through phase	All identified security issues resolved; penetration test findings addressed
CP-22: Compliance closeout (Ethiopia)	60% through phase	All Ethiopia regulatory requirements documented and satisfied

CP-23: Production release dress rehearsal	75% through phase	Full production deployment simulated end-to-end including rollback
CP-24: Production release signoff	95% through phase	Founding team signs off on production readiness across all dimensions

Gate Criteria for Production Launch:

- [ ] Current prototype scope is production-safe across web and patient mobile
- [ ] Trained staff in place with documented schedules and escalation paths
- [ ] Fallback operations documented and rehearsed
- [ ] Compliance requirements met for launch geography (Ethiopia)
- [ ] Kenya launch plan ready (even if Kenya launch is post-production)
- [ ] Release governance fully operational with rollback tested
- [ ] Monitoring and alerting covering all critical paths
- [ ] Support schedule covering operating hours with documented handoff procedures
- [ ] Revenue at production pricing for at least 1 month
- [ ] Monthly burn rate sustainable for 6+ months (per params.md success metric)
- [ ] LTV:CAC tracking in place with initial data

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4. Timeline Matrix Across All 21 Paths

4.1 Calculation Methodology

The timeline for each path is derived from the total effort required, team capacity at each investment level, and the sequential dependencies between phases.

Total effort to be scheduled:

Category	Hours	Notes
Web platform hardening	745	Runs Pre-Pilot through Pilot 2; senior-rate work
Android APK	480	Parallel workstream on dedicated sub-team; Pre-Pilot through Pilot 2
QA, DevOps, security	220	Spread across all phases with concentration in Pilot 2 and Prod Readiness
Parallel ops workstreams	750	Consumed by founders/founding team; runs Pre-Pilot through Prod Readiness
Total	**2,195**	Per params.md Section 17.3

Monthly capacity by investment level (per params.md Section 2.4):

Level	Technical Hours/Mo	Effective Split: Web	Effective Split: APK	Effective Split: QA/DevOps
L1	320	~120-140	~100-120	~30-40
L2	480	~175-200	~150-170	~50-60
L3	640	~230-260	~200-220	~70-80

Operational workstream throughput is bounded by founder/founding-team availability, not developer capacity. At all levels, ops workstream throughput is approximately 40-80 hours/month (constrained by the 3 core team members splitting time across ops, strategy, investor relations, and product oversight).

Key constraint: Phase advancement requires both technical gates AND operational gates to be met. The binding constraint is whichever takes longer.

4.2 Phase Duration Logic

Pre-Pilot duration is driven by: (a) web hardening to reach core stability (~~200-280 hours), (b) APK sprint 0 and core scaffolding (~~100-160 hours), (c) operational workstream drafting (~80-120 hours), and (d) 2 rehearsal cycles requiring calendar time regardless of team size. Minimum calendar time: 3 months (L3), maximum: 6 months (L1).

Pilot 1 duration is relatively fixed at 2-3 months because it is bounded by the need to run a live pilot for at least 1 month with real patients. Calendar time dominates over effort here.

Pilot 2 duration is the most variable phase. It absorbs the remaining technical work, the staffing ladder scale-up, the analytics build-out, and the Kenya prep. At L1 without investor capital, this phase stretches to 13 months. At L3 with full funding, it compresses to 5 months.

Production Readiness duration scales with the depth of hardening required and the maturity of operational systems entering the phase. Paths that formalized operations earlier (L3) need less production readiness time.

4.3 Full 21-Path Timeline Matrix

#	Scenario	Family	Pre-Pilot	Pilot 1	Pilot 2	Prod Readiness	Total Months
1	S1-L1	Self-funded, bootstrapped	M0-M6	M6-M9	M9-M22	M22-M34	**34**
2	S1-L2	Self-funded, ideal	M0-M5	M5-M7	M7-M16	M16-M26	**26**
3	S1-L3	Self-funded, fully funded	M0-M4	M4-M6	M6-M12	M12-M20	**20**
4	S2-O1-I1	Inv. during transition, bootstrap/bootstrap	M0-M6	M6-M9	M9-M18	M18-M30	**30**
5	S2-O1-I2	Inv. during transition, bootstrap/ideal	M0-M6	M6-M9	M9-M17	M17-M27	**27**
6	S2-O1-I3	Inv. during transition, bootstrap/full	M0-M6	M6-M9	M9-M15	M15-M24	**24**
7	S2-O2-I1	Inv. during transition, ideal/bootstrap	M0-M5	M5-M7	M7-M16	M16-M27	**27**
8	S2-O2-I2	Inv. during transition, ideal/ideal	M0-M5	M5-M7	M7-M14	M14-M24	**24**

9	S2-O2-I3	Inv. during transition, ideal/full	M0-M5	M5-M7	M7-M13	M13-M21	**21**
10	S2-O3-I1	Inv. during transition, full/bootstrap	M0-M4	M4-M6	M6-M14	M14-M25	**25**
11	S2-O3-I2	Inv. during transition, full/ideal	M0-M4	M4-M6	M6-M12	M12-M22	**22**
12	S2-O3-I3	Inv. during transition, full/full	M0-M4	M4-M6	M6-M11	M11-M19	**19**
13	S3-O1-I1	Inv. after Pilot 1, bootstrap/bootstrap	M0-M5	M5-M8	M8-M16	M16-M26	**26**
14	S3-O1-I2	Inv. after Pilot 1, bootstrap/ideal	M0-M5	M5-M8	M8-M14	M14-M22	**22**
15	S3-O1-I3	Inv. after Pilot 1, bootstrap/full	M0-M5	M5-M8	M8-M13	M13-M20	**20**
16	S3-O2-I1	Inv. after Pilot 1, ideal/bootstrap	M0-M4	M4-M6	M6-M14	M14-M24	**24**
17	S3-O2-I2	Inv. after Pilot 1, ideal/ideal	M0-M4	M4-M6	M6-M12	M12-M20	**20**
18	S3-O2-I3	Inv. after Pilot 1, ideal/full	M0-M4	M4-M6	M6-M11	M11-M18	**18**
19	S3-O3-I1	Inv. after Pilot 1, full/bootstrap	M0-M3	M3-M5	M5-M12	M12-M22	**22**
20	S3-O3-I2	Inv. after Pilot 1, full/ideal	M0-M3	M3-M5	M5-M10	M10-M18	**18**
21	S3-O3-I3	Inv. after Pilot 1, full/full	M0-M3	M3-M5	M5-M9	M9-M16	**16**

4.4 Summary Statistics

Metric	Value
Shortest path	S3-O3-I3: **16 months**
Longest path	S1-L1: **34 months**
Median path	**22 months**
Mean path	**23.0 months**
Primary recommended (S3-O2-I2)	**20 months**

4.5 Phase Duration Ranges Across All Paths

Phase	Shortest	Longest	Median	Notes
Pre-Pilot	3 months	6 months	4-5 months	Bounded by rehearsal cycles and APK sprint 0
Pilot 1	2 months	3 months	2 months	Bounded by minimum live-pilot duration
Pilot 2	4 months	13 months	7 months	Most variable; absorbs funding timing impact
Production Readiness	7 months	12 months	8-10 months	Scales with ops maturity entering the phase

4.6 Effort Consumption by Phase (Expected Case, L2 Baseline)

Phase	Web Hours	APK Hours	QA/DevOps Hours	Ops Hours	Total Hours	Duration
Pre-Pilot	240	130	30	100	500	4-5 months
Pilot 1	175	220	40	120	555	2 months
Pilot 2	250	130	100	300	780	6-8 months
Prod Readiness	80	0	50	230	360	8 months
Total	**745**	**480**	**220**	**750**	**2,195**	**20-23 months**

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5. How to Interpret the Timing Differences

5.1 Why Level 1 Paths Stretch

Level 1 paths (bootstrapped, ~320 hrs/mo technical capacity) do not stretch only because "code is slower." They stretch because of compounding constraints:

Founder time fragmentation. At L1, the 3 core team members are the primary contributors to both technical work and operational workstreams. With only ~~264 hours/month of core team capacity and ~572 hours of developer capacity at reduced utilization (~~56%), the founders must context-switch constantly between code review, SOP writing, partner meetings, investor outreach, and product decisions. Each switch carries a productivity tax.

Manual operations stay informal longer. Without the budget to formalize SOPs early or hire ops staff ahead of schedule, L1 paths run Pilot 1 with informal verbal procedures. This works for 100 users but creates bottlenecks at 500+, forcing a pause to document what should have been documented earlier.

Staffing formalization happens later. The employed ops staff ladder (doctors, admin, tech support) starts at Pilot 1 in all scenarios, but at L1, the onboarding is less structured and the staff have less documentation to work from, leading to slower ramp-up and more founder intervention.

Partner onboarding becomes a bottleneck. At L1, the partner onboarding kit is often incomplete or ad hoc, meaning each new clinic requires custom hand-holding rather than a repeatable process. This slows Pilot 2 scaling.

Quantified impact: L1 adds 8-14 months versus L3 across the full timeline. The majority of this stretch (60-70%) occurs in Pilot 2 and Production Readiness, not in Pre-Pilot.

5.2 Why Level 3 Paths Shorten

Level 3 paths (~640 hrs/mo technical capacity) compress the timeline through several mechanisms:

More parallelism is safe. With enough developer bandwidth, the web team and APK team can work truly independently without founders acting as shared bottlenecks. The web DevOps engineer and full-stack developer can handle hardening while the Android sub-team builds the APK, while the flex developer bridges gaps — all without contention.

Operations formalize earlier. At L3, there is enough founding-team bandwidth to write SOPs, train staff, and build partner kits concurrently with development. This means the operational gates are met earlier, rather than becoming the binding constraint after technical work completes.

Hardening and launch prep do not queue behind features. At L1-L2, security hardening, load testing, and production readiness tasks often wait in the backlog behind feature work. At L3, these can be tackled in parallel, eliminating the sequential bottleneck.

Quantified impact: L3 saves 8-14 months versus L1. The savings are distributed across all phases but are most pronounced in Pilot 2 (4-8 months shorter) and Production Readiness (2-4 months shorter).

5.3 Why Timelines Do Not Collapse to Half Despite Reduced Web Hours

This is the most important interpretive point for investors evaluating the plan.

The v2 parameter set reduced the web platform estimate from v1's broad range to a tight 745-hour expected case. This is a real improvement. But the total effort budget is 2,195 hours, of which web is only 34%. The remaining 66% is:

Non-Web Category	Hours	% of Total	Compressible by More Devs?
Android APK	480	22%	Partially (dedicated team, but parallelism has limits)
QA/DevOps/Security	220	10%	Partially (requires web + APK to be testable first)
Parallel ops workstreams	750	34%	No (bounded by founder/founding-team bandwidth and calendar time)

The parallel ops workstreams are the key. They are bounded by:

1. **Founder availability** — not developer headcount
2. **Calendar time** — doctor training requires actual doctors to be hired, trained, and observed over weeks; SOPs require real operational data to iterate on; compliance research requires regulatory engagement cycles
3. **Sequential dependencies** — you cannot finalize a tech support runbook before the tech support team exists (Pilot 1); you cannot calibrate staffing before you have staffing data (Pilot 2)

This is why doubling developer capacity from L1 to L2 does not halve the timeline. It compresses the technical phases but leaves the operational floor intact.

5.4 Impact of Investor Timing (S2 vs S3)

The difference between S2 (investor during Pilot 1-to-Pilot 2 transition) and S3 (investor after Pilot 1) is typically 2-4 months.

S3 paths are slightly shorter because:

- Earlier capital injection enables earlier staffing ladder activation
- Pilot 2 begins with full resources rather than ramping up mid-phase
- The investor confidence signal (earlier commitment) reduces founder time spent on fundraising

However, S2 paths have a strategic advantage: the company has more operating proof by the time capital arrives, which may enable better terms and lower dilution.

5.5 The Investor Level Matters More Than Investor Timing

Across the matrix, moving from I1 to I3 (within the same scenario) typically saves 4-8 months. Moving from S2 to S3 (within the same investment level) saves only 2-4 months. This means the quality and size of the investment matters more than its timing, within reasonable bounds.

Comparison	Average Months Saved
I1 to I2 (same scenario)	3-4 months
I2 to I3 (same scenario)	2-3 months
S2 to S3 (same investment level)	2-3 months
O1 to O2 (same scenario)	3-4 months
O2 to O3 (same scenario)	2-3 months

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6. Recommended Timeline View

6.1 Primary Recommendation: S3-O2-I2

Attribute	Value
Scenario	S3 — Investor after Pilot 1
Our spend level	O2 — Ideal (480 hrs/mo technical capacity)
Investor level	I2 — Ideal
Total duration	**20 months**
Pre-Pilot	M0-M4 (4 months)
Pilot 1	M4-M6 (2 months)
Pilot 2	M6-M12 (6 months)
Production Readiness	M12-M20 (8 months)

Rationale: This path balances three critical factors:

1. **Proof before capital.** By self-funding through Pilot 1 at ideal capacity, the company demonstrates live-user viability, callback discipline, and first revenue before asking for outside capital. This creates a stronger negotiating position and reduces dilution risk.

2. **Financing timing aligns with the cost escalation curve.** The employed ops staff ladder ($2,482/mo at Pilot 1, scaling to $6,542/mo at Production Readiness) and infrastructure costs ($270-520/mo at Pilot 1, scaling to $800-1,500/mo at Production) create a cost inflection at Pilot 2. Investor capital arriving at exactly this point prevents the founders from bearing the scaling burden alone.

3. **Operational maturity has time to develop.** 20 months provides enough calendar time for the 750 hours of parallel operational workstreams to be completed without rushing. Doctor training gets real iteration cycles. Admin SOPs get real data. The staffing ladder scales with genuine operational learning.

6.2 Fallback: S2-O2-I2

Attribute	Value
Scenario	S2 — Investor during Pilot 1-to-Pilot 2 transition
Our spend level	O2 — Ideal
Investor level	I2 — Ideal
Total duration	**24 months**
Pre-Pilot	M0-M5 (5 months)
Pilot 1	M5-M7 (2 months)
Pilot 2	M7-M14 (7 months)
Production Readiness	M14-M24 (10 months)

Rationale: If outside capital lands later than hoped (during the transition rather than immediately after Pilot 1), this path preserves operating discipline. The 4 additional months versus the primary path are absorbed primarily in Pilot 2 (+1 month) and Production Readiness (+2 months), where the delayed funding means slower staffing ladder activation and more sequential hardening work.

This is the "plan B" that does not require changing strategy — only adjusting the timeline expectation. The same work gets done; it just takes longer because the cost scaling starts later.

6.3 Upside: S3-O2-I3

Attribute	Value
Scenario	S3 — Investor after Pilot 1
Our spend level	O2 — Ideal
Investor level	I3 — Fully Funded
Total duration	**18 months**
Pre-Pilot	M0-M4 (4 months)
Pilot 1	M4-M6 (2 months)
Pilot 2	M6-M11 (5 months)
Production Readiness	M11-M18 (7 months)

Rationale: If an investor provides fully-funded-level capital after Pilot 1, the company can compress Pilot 2 by 1 month and Production Readiness by 1 month. The savings come from:

- Faster staffing ladder activation (full complement hired and trained earlier)
- More parallel technical work (hardening, analytics, and Kenya prep done concurrently)
- Reduced founder fundraising burden (more time for operational oversight)

This path is fast but still defensible. The 18-month timeline preserves all four phases and all gate criteria. It does not skip operational maturation — it accelerates it through better resourcing.

Caution: This path requires investor quality, not just investor quantity. "Fully funded" means the investor understands the healthcare operations model and does not pressure the company to skip operational readiness in favor of faster user growth. A poorly aligned L3 investor could be worse than a well-aligned L2 investor.

6.4 Recommended View Summary

Path	Scenario	Duration	Monthly Burn at Peak	Key Risk
Primary	S3-O2-I2	**20 months**	~$21,000-24,000	Investor timing must align with Pilot 1 completion
Fallback	S2-O2-I2	**24 months**	~$19,000-22,000	Longer self-funded runway increases founder cash burden
Upside	S3-O2-I3	**18 months**	~$25,000-30,000	Requires high-quality, mission-aligned investor

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Appendix A: Key Differences from v1 Timeline

Parameter	v1 Value	v2 Value	Impact on Timeline
Team model	"Small team of 3" abstraction	12 part-time people with explicit capacity	Phase durations now grounded in real throughput math
Web effort	682-1,413 hrs (3-point)	745 hrs (team estimate)	Pre-Pilot and Pilot 2 technical work tightened by ~15-20%

APK effort	440-640 hrs (range)	480 hrs (contracted)	APK milestone timing now deterministic
Ops workstreams	Mentioned but not quantified	750 hours explicitly modeled	Explains why timeline floor exists regardless of dev speed
Employed staff	Not explicitly modeled	Ladder starting at Pilot 1 ($2,482/mo)	Pilot 1 gate now includes staffing readiness
Ops infrastructure	Not modeled	Phones, VoIP, call center software	Pre-Pilot and Pilot 1 have real setup requirements
Capacity levels	Implied	L1=320, L2=480, L3=640 hrs/mo	Phase durations derived from explicit math, not estimates
Total effort	Not summed	2,195 hours	Single number enables cross-path comparison

Appendix B: Glossary of Timeline Notation

Term	Meaning
M0	Month zero; project start
M0-M4	Phase runs from month 0 through month 4
S1/S2/S3	Scenario 1 (self-funded), Scenario 2 (investor during transition), Scenario 3 (investor after Pilot 1)
O1/O2/O3	Our spend level: bootstrapped / ideal / fully funded
I1/I2/I3	Investor spend level: bootstrapped / ideal / fully funded
L1/L2/L3	Investment level (used for S1 where there is only one spend level)
CP-XX	Checkpoint number (internal minor gate)
W1-W9	Parallel operational workstream number

End of 01-timeline.md — this document is the phase-order source of truth for the Health Hub v2 business plan. All downstream documents reference this file for phase timing and gate criteria.